

Assignment 2 – Case Study Lab

Stage 2 Lab Activities

SmartCare Requirements Engineering

AI OFF -> AI ON -> VERIFY | 1 hour

Learning objectives

- Analyse the SmartCare client brief.
- Identify stakeholders and scope.
- Write functional and non-functional requirements.
- Develop user stories and Given-When-Then acceptance criteria.
- Use AI to critique requirements without allowing it to invent stakeholder needs.
- Produce SmartCare Requirements Specification v1.0.

Part A - Client Brief: AI OFF

SmartCare uses spreadsheets and paper records. Staff report duplicate bookings, difficulty finding patient information, inconsistent appointment status and limited appointment history. Management wants a small, maintainable patient, practitioner and appointment system.

Part B - Stakeholders and Scope: AI OFF

Identify at least four stakeholders. Create In Scope and Out of Scope lists. Label uncertain features as provisional rather than confirmed.

Stakeholders:

1. Patients: Need proper, easy booking system without duplicate bookings
2. Practitioners: Need to view patient history, schedules and appointment notes
3. Management: Utilizing, managing system and increasing reliability
4. IT Support: Responsible for maintaining the system, edit data.

In Scope:

1. Maintaining patient's details
2. Booking an appointment, rescheduling and cancelling an appointment
3. Preventing duplicate booking of appointment
4. Practitioner record management

Out of scope:

1. SMS or email remainder
2. Electronically prescribing a medication
3. Insurance claims handling

4. Multi clinic facility

Provisional:

1. Automated appointment reminder through SMS or emails
2. Management for cancelations of appointment
3. Role based access to the system

Part C - Functional Requirements: AI OFF

Write 8-12 numbered functional requirements using FR-01, FR-02 and so on. Each should describe one observable capability.

FR-01: System should allow staff to reschedule an appointment

FR-02: System should allow staff to update patient details, information

FR-03: It should allow staffs to update existing information about the patients

FR-04: The system should block duplicate booking

FR-05: The system should allow users to update the practitioner's availability

FR-06: The system should allow staff to cancel an appointment

FR-07: The system should display a practitioner's weekly schedules

FR-08: The system should be able to filters old appointments

Part D - Non-Functional Requirements: AI OFF

Write 4-6 numbered non-functional requirements covering appropriate qualities such as reliability, maintainability, usability, data integrity or testability.

NFR-01: Reliability: The system should be available all the time except for any update

NFR-02: Maintenance: The system must be upgradeable without hampering any existing service unless modification is required.

NFR-03: Data Security: Patients records should be accessible only by the verified staff.

NFR-04: Data Integrity: The system should avoid double bookings

Part E - User Stories and Acceptance Criteria: AI OFF

Write 4-6 user stories. For at least three, create Given-When-Then acceptance criteria including one negative or failure scenario.

US-0: As a receptionist, I want to book an appointment for a patient to a specific doctor.

US-02: As a patient, I want to view my appointment and receive a reminder

US-03: As a receptionist, I want to cancel an appointment

US-04: As a practitioner, I want to view my daily schedule

US-05: As a receptionist, I want to make a change in the appointment when needed

Given_When_Then:

Story 1: Success

Given a practitioner has an available time on a date

When the receptionist books the appointment for a patient

Then the system saves the appointment and displays a confirmation.

Story 2: failure

Given an appointment has already been completed

When the receptionist tries to cancel the appointment

Then the system prevents the cancellation

Story 4: Success

Given a practitioner has appointments booked for the current day,

When the practitioner opens their daily schedule view,

Then the system displays all of that day's appointments in chronological order, including patient name and time.

Part F - AI Requirements Review: AI ON

Prompt: Act as a software requirements reviewer. Review the SmartCare requirements for ambiguity, inconsistency, missing clarification questions and testability. Do NOT invent new client requirements. For every suggestion, state whether it is based on evidence or is only a question/assumption requiring validation.

Ambiguity

- [Evidence] FR-07 specifies "weekly schedules" while Story 4's acceptance criteria describes a *daily* schedule view. It's unclear whether both views are required or which one is the actual requirement.
- [Assumption] FR-04 ("block duplicate booking") doesn't specify what "duplicate" means — the same patient booked twice, the same practitioner double-booked, or an identical repeated entry. These imply different rules.
- [Assumption] FR-08 ("filter old appointments") — does "old" mean past-dated, or status = completed/cancelled? Not defined, and no filter criteria are given.

Inconsistency

- [Evidence] Two requirements are both effectively "FR-01": one for rescheduling ("FR-01") and one for updating patient details ("FR_01", using an underscore instead of a hyphen). This creates a duplicate/clashing ID, and FR-03 then separately repeats the same "update patient information" capability — so that capability appears twice under two different numbers.
- [Evidence] FR-5 is missing its leading zero (should be FR-05), inconsistent with the FR-01/03/04/06/07/08 format.

- [Evidence] Story 2's cancellation rule ("can't cancel a completed appointment") is not reflected in FR-06 ("system should allow staff to cancel an appointment") — the FR carries no restriction, so the negative scenario in Part E has no corresponding functional rule to test against.
- [Evidence] NFR "Data Integrity" ("avoid double bookings") overlaps with FR-04 ("block duplicate booking"), but the two are worded differently and it isn't clear whether they refer to the same rule (practitioner double-booked) or two separate rules (also covering duplicate patient bookings).

Missing Clarifications

- [Assumption] Who is permitted to update practitioner availability (FR-5) — the practitioner, reception staff, or both?
- [Assumption] Story 2 (view appointment + "receive a reminder") assumes automated reminders exist, but SMS/email reminders were listed as provisional in Part B, not confirmed. This needs client sign-off before being treated as firm scope.
- [Assumption] No FR covers searching for or viewing a patient's appointment history, despite the original brief citing "difficulty finding patient information" as a core pain point and practitioners needing to view patient history. Worth confirming whether this is an intentional omission or a gap.
- [Assumption] Story 5 ("make a change in the appointment when needed") has no acceptance criteria and overlaps conceptually with Story 1 (rescheduling) — is it a distinct capability or a duplicate of FR-01?

Testability

- [Evidence] NFR "Reliability" ("available all the time except for any update") has no measurable target (e.g., uptime percentage, defined maintenance window), so it cannot be objectively verified as written.
- [Evidence] NFR "Maintenance" ("upgradeable without hampering any existing service unless modification is required") is circularly worded — it doesn't specify a pass/fail condition a tester could check against.
- [Assumption] FR-08's "filter old appointments" has no defined expected output to verify against until "old" is defined.

Part G - VERIFY the AI Review

Classify each significant AI suggestion as Accepted, Modified, Rejected, or Unverified. Explain the evidence used.

Verification of AI Review

	Suggestion	Classification	Evidence
1	Availability logic	Accepted	FRs don't cross-reference each other — confirmed by re-reading.
2	FR-2 duplicate, and FR-03 repeating the same capability	Accepted	Confirmed directly from the numbering as submitted FR-02 and duplicates FR-03's content
3	FR-07 ("weekly schedules") conflicts with Story 4's daily schedule view	Accepted	Confirmed by reading that FR-07 and Story 4 describe different time windows with no explanation of which applies.
4	Status-transition rules is only in user stories, not on the FRs	Accepted	FR-08/09 imposes no restriction
5	Who edits practitioner availability (FR-5) is undefined	Accepted	FR-5 says "users" only, no role specified.
6	NFR "Data Integrity" overlaps with FR-04	Modified	Real overlap exists
7	Practitioner availability edit is not defined	Accepted	FR-04 says "staff" only, no role distinction.
8	Story 2's reminder assumes automated reminders, contradicting "provisional" status in Part B	Accepted	Part B lists SMS/email reminders as provisional, not confirmed
9	Duplication of patient record handling is missing	Accepted	No FR/NFR covers this; genuine gap.
10	No FR covers patient history search/viewing	Accepted	Confirmed gap

Part H - Finalise SmartCare v0.2

Submit stakeholder analysis, scope, 8-12 FRs, 4-6 NFRs, 4-6 user stories, acceptance criteria, assumptions/open questions and selected AI review evidence.

Stakeholder analysis:

1. Patients: need accurate records, no duplicate, cancelling appointments
2. Practitioners: need to view schedules, patient history
3. Management: needs reporting, increasing reliability
4. IT support: Book schedule, cancel and edit data

In scope:

1. Managing patient data
2. Managing practitioner data

3. Appointment booking, rescheduling, cancellation
4. Duplicate/overlapping booking prevention

Out of scope:

1. Billing processing
2. Auto SMS or email reminder
3. Self service by the patient
4. Electronic prescribing

Functional requirements:

1. Rescheduling appointment
2. Cancelling an appointment
3. Creating patient record
4. Updating patient data
5. Prevention of duplicate booking
6. Practitioners daily schedule
7. Filter completed tasks
8. Displaying patient history when searched for

Non-functional requirements:

1. Patient records accessible only to authenticated staff
2. Staff shall be able to book an appointment
3. Staff shall be able to cancel an appointment
4. System shall prevent double-booking

User stories:

1. As a receptionist, I want to book an appointment for a patient to a specific doctor.
2. As a patient, I want to view my appointment and receive a reminder
3. As a receptionist, I want to cancel an appointment
4. As a practitioner, I want to view my daily schedule

Selected AI review evidence:

1. [Evidence] Two requirements are both effectively "FR-01": one for rescheduling ("FR-01") and one for updating patient details ("FR_01", using an underscore instead of a hyphen). This creates a duplicate/clashing ID, and FR-03 then separately repeats the same "update patient information" capability — so that capability appears twice under two different numbers.
2. [Evidence] FR-07 specifies "weekly schedules" while Story 4's acceptance criteria describes a *daily* schedule view. It's unclear whether both views are required or which one is the actual requirement.

Reflection

In 150-250 words: What did AI notice that you missed? What did AI invent or overreach on? Which requirement changed after review? Why must requirements have evidence?

AI found some interesting mistakes which were not done intentionally. The AI did not invent new stakeholders or new events but it mentioned one suggestion that the NFR data integrity was overlapping with FR 4, which assumed that as an inconsistency. And so I classified it as modified instead of Accepted. Again, FR-06 changed: it originally had no restriction on cancelling completed appointments, so I added a status-transition rule to match the negative scenario already described in Story 2.